

Gradient-Product-Balanced Initialization

A Per-Layer Variance Schedule for Row-Centered ReLU Networks

Itay Ofer

April 2026

Contents

1	Goal and Motivation	2
2	The Forward and Backward Gains for Row-Centered Weights	2
2.1	Notation	2
2.2	Forward gain derivation	3
2.3	Backward gain derivation	3
2.4	The coupling formula	3
2.5	Summary of existing variants	4
3	The Product-Balanced Initializer	4
3.1	Derivation	4
3.2	Resulting gains	4
3.3	The initialization procedure	4
4	Empirical Validation (50 Hidden Layers)	4
4.1	Setup	4
4.2	Gain verification	5
4.3	Forward gain per layer	5
4.4	Backward gain per layer	6
4.5	Gain product per layer	6
5	Geometry Preservation (k-NN Accuracy)	6
5.1	Setup	6
5.2	Results	7
6	Per-Layer Variance Scaling: The η-Schedule	7
6.1	Derivation from BP4	7
6.2	The η -parameterization	8
7	The Pareto Frontier	8
7.1	The fundamental trade-off	8
7.2	Numerical consequences	9
7.3	Depth-adaptive η	9
7.4	Varying widths	10
8	Conclusion	10
8.1	Results	10
8.2	Implications	10

1 Goal and Motivation

We seek an initialization scheme for deep ReLU networks with row-centered weights that simultaneously preserves geometry (class separability under deep projections) and maintains gradient stability (uniform gradient magnitudes across layers).

The fourth backpropagation equation (BP4) gives the gradient of the loss $\mathcal{C}$ with respect to the weight matrix $W^{(\ell)}$ at layer ℓ :

$$\frac{\partial \mathcal{C}}{\partial W^{(\ell)}} = \frac{1}{n} (\Delta^{(\ell)})^\top A^{(\ell-1)}, \quad (1)$$

where $A^{(\ell-1)} \in \mathbb{R}^{n \times d_{\ell-1}}$ is the activation matrix at layer $\ell-1$ (the input to layer ℓ) and $\Delta^{(\ell)} \in \mathbb{R}^{n \times d_\ell}$ is the error signal matrix at layer ℓ . Taking the Frobenius norm:

$$\left\| \frac{\partial \mathcal{C}}{\partial W^{(\ell)}} \right\|_F \propto \text{rms}(A^{(\ell-1)}) \cdot \text{rms}(\Delta^{(\ell)}). \quad (2)$$

For the gradient to be uniform across layers, we need the product $\text{rms}(A^{(\ell-1)}) \cdot \text{rms}(\Delta^{(\ell)})$ to be approximately constant in ℓ .

This product is governed by two quantities:

- The **forward gain** at layer ℓ : $g_{\text{fwd}}^{(\ell)} = \frac{\text{rms}(A^{(\ell)})}{\text{rms}(A^{(\ell-1)})}$.
- The **backward gain** at layer ℓ : $g_{\text{bwd}}^{(\ell)} = \frac{\text{rms}(\Delta^{(\ell)})}{\text{rms}(\Delta^{(\ell+1)})}$.

We define the **gradient ratio** as the ratio between the largest and smallest gradient norms across hidden layers:

$$\text{Gradient ratio} = \frac{\max_\ell \|\partial \mathcal{C} / \partial W^{(\ell)}\|_F}{\min_\ell \|\partial \mathcal{C} / \partial W^{(\ell)}\|_F}. \quad (3)$$

A gradient ratio of 1 means perfectly uniform gradients; a ratio of $100\times$ means some layers receive $100\times$ larger gradients than others.

2 The Forward and Backward Gains for Row-Centered Weights

We now derive the forward and backward gain formulas for a row-centered layer. These results were established in the main report (Chapters 3–4); we include the full derivation here for completeness.

2.1 Notation

Consider a single layer with weight matrix $W \in \mathbb{R}^{d \times d}$ (constant width for simplicity; the varying-width case is discussed in Section 7.4). The layer computes:

$$\mathbf{h}^{(\ell)} = \text{ReLU}(W^{(\ell)} \mathbf{h}^{(\ell-1)}).$$

Row centering means $\sum_j W_{ij} = 0$ for every row i . The per-element variance is $\text{Var}(W_{ij}) = v/d$, where $v > 0$ is a *variance scale factor*.

We define the **variance factor**:

$$s = \sqrt{\frac{v}{2}}, \quad (4)$$

so that $s = 1$ corresponds to standard He initialization ($\text{Var}(W_{ij}) = 2/d$), and other values of s correspond to scaled variants. This parameterization is convenient because the He initialization is designed so that the forward gain equals 1 when there is *no* row centering.

2.2 Forward gain derivation

For a standard He-initialized layer (no centering), the expected output energy is:

$$\mathbb{E}[(a^{(\ell)})^2] = \underbrace{\text{Var}(W) \cdot d}_{=(2/d) \cdot d=2} \cdot \underbrace{\frac{1}{2}}_{\text{ReLU survival}} \cdot \mathbb{E}[(a^{(\ell-1)})^2] = \mathbb{E}[(a^{(\ell-1)})^2]. \quad (5)$$

The key step: the factor $\mathbb{E}[(a^{(\ell-1)})^2]$ appears because He weights use the *full* second moment of their input.

For **row-centered** weights, $\sum_j W_{ij} = 0$ implies (Proposition 3.3 in the main report) that the neuron output depends only on the deviations $a_j - \bar{a}$:

$$z_i = \sum_j W_{ij} a_j = \sum_j W_{ij} (a_j - \bar{a}).$$

The effective input energy is therefore $\text{Var}(a)$ instead of $\mathbb{E}[a^2]$. For post-ReLU activations with pre-activation $z \sim \mathcal{N}(0, \sigma^2)$:

$$\mathbb{E}[a] = \frac{\sigma}{\sqrt{2\pi}}, \quad \mathbb{E}[a^2] = \frac{\sigma^2}{2}, \quad \text{Var}(a) = \mathbb{E}[a^2] - (\mathbb{E}[a])^2 = \frac{\sigma^2}{2} \left(1 - \frac{1}{\pi}\right) = \frac{\sigma^2(\pi - 1)}{2\pi}. \quad (6)$$

The fraction of energy accessible to row-centered weights is:

$$\frac{\text{Var}(a)}{\mathbb{E}[a^2]} = \frac{\sigma^2(\pi - 1)/(2\pi)}{\sigma^2/2} = \frac{\pi - 1}{\pi} \approx 0.6817. \quad (7)$$

This ratio is *independent of σ* — it is a universal property of the ReLU activation.

With variance $\text{Var}(W_{ij}) = v/d$ and defining $s = \sqrt{v/2}$, the output energy becomes:

$$\mathbb{E}[(a^{(\ell)})^2] = \underbrace{s^2}_{=v/2} \cdot \underbrace{\frac{\pi - 1}{\pi}}_{\text{centering ratio}} \cdot \mathbb{E}[(a^{(\ell-1)})^2].$$

Taking square roots to get the forward gain (ratio of RMS values):

$$\boxed{g_{\text{fwd}} = r \cdot s, \quad \text{where } r = \sqrt{\frac{\pi - 1}{\pi}} \approx 0.826.} \quad (8)$$

The factor r is the **structural centering factor**: it is a consequence of row centering + ReLU and cannot be changed by adjusting the variance.

2.3 Backward gain derivation

The backward pass propagates the error signal via $\delta^{(\ell)} = (W^{(\ell+1)})^\top \delta^{(\ell+1)} \odot \sigma'(z^{(\ell)})$, where $\sigma'(z^{(\ell)})$ is the ReLU derivative (a binary mask: 1 where $z > 0$, else 0).

Row centering of W makes $W^\top$ column-centered: $\sum_i W_{ij} = 0$ for each column j . The error signal δ has near-zero mean (it contains both positive and negative values), so $\text{Var}(\delta)/\mathbb{E}[\delta^2] \approx 1$. Unlike post-ReLU activations, there is essentially no “DC component” to lose.

Therefore, the backward gain depends only on the weight magnitude:

$$\boxed{g_{\text{bwd}} = s.} \quad (9)$$

2.4 The coupling formula

Dividing (8) by (9):

$$\frac{g_{\text{fwd}}}{g_{\text{bwd}}} = r \approx 0.826, \quad \text{always, regardless of } s. \quad (10)$$

This is the fundamental constraint: no single scalar variance can make both gains equal to 1 simultaneously.

2.5 Summary of existing variants

Table 1: Existing row-centered initializers and their gain profiles.

Name	s	g_{fwd}	g_{bwd}	Product
Variance-adjusted (He variance)	1.000	0.826	1.000	0.826
Forward-balanced	1.211	1.000	1.211	1.211
Product-balanced (new)	1.101	0.909	1.101	1.000

3 The Product-Balanced Initializer

3.1 Derivation

We seek s^* such that $g_{\text{fwd}} \cdot g_{\text{bwd}} = 1$:

$$\begin{aligned}
 r \cdot s^2 &= 1 \\
 s^2 &= \frac{1}{r} = \sqrt{\frac{\pi}{\pi-1}} \\
 s^* &= \left(\frac{\pi}{\pi-1} \right)^{1/4} \approx 1.1006.
 \end{aligned} \tag{11}$$

The corresponding weight variance is:

$$\text{Var}(W_{ij}) = \frac{2 s^{*2}}{d} = \frac{2\sqrt{\pi/(\pi-1)}}{d} \approx \frac{2.422}{d}. \tag{12}$$

3.2 Resulting gains

$$g_{\text{fwd}}^* = r \cdot s^* = \left(\frac{\pi-1}{\pi} \right)^{1/4} \approx 0.9089, \tag{13}$$

$$g_{\text{bwd}}^* = s^* = \left(\frac{\pi}{\pi-1} \right)^{1/4} \approx 1.1006, \tag{14}$$

$$g_{\text{fwd}}^* \cdot g_{\text{bwd}}^* = 1.0 \quad (\text{exact}). \tag{15}$$

3.3 The initialization procedure

Definition 3.1 (Product-balanced row-centered initialization). Given a layer with fan-in d :

1. Compute $v^* = 2\sqrt{\pi/(\pi-1)} \approx 2.4224$.
2. Set $\sigma = \sqrt{v^*/d}$.
3. Sample $W_{ij} \sim \mathcal{N}(0, \sigma^2)$ for all i, j .
4. Row-center: $W_{ij} \leftarrow W_{ij} - \frac{1}{d} \sum_k W_{ik}$.
5. Rescale each row to restore the target standard deviation: $\mathbf{w}_i \leftarrow \mathbf{w}_i \cdot \sigma / \text{std}(\mathbf{w}_i)$.
6. Set bias to zero.

4 Empirical Validation (50 Hidden Layers)

4.1 Setup

Fashion-MNIST, 2000 samples, architecture $784 \rightarrow [784]^{50} \rightarrow 1$. We compare three initializers:

- **He**: standard Kaiming initialization (baseline).

- **RC forward-balanced**: row-centered with $g_{\text{fwd}} = 1$, $g_{\text{bwd}} \approx 1.21$.
- **RC product-balanced**: row-centered with $g_{\text{fwd}} \cdot g_{\text{bwd}} = 1$.

4.2 Gain verification

Table 2: Per-layer gain comparison, 50 hidden layers, width 784.

Initializer	Med g_{fwd}	Med g_{bwd}	Product	Grad ratio	Grad CV
He (baseline)	0.993	0.997	0.990	$7.8\times$	0.522
RC fwd-balanced	1.000	1.209	1.210	$1901\times$	1.903
RC product-balanced	0.909	1.099	0.999	$1902\times$	1.906

The empirical gains match the theory to within 0.2%. The gradient ratio is $\sim 1900\times$ for both RC variants, confirming that *uniform* variance alone does not fix gradient non-uniformity.

4.3 Forward gain per layer

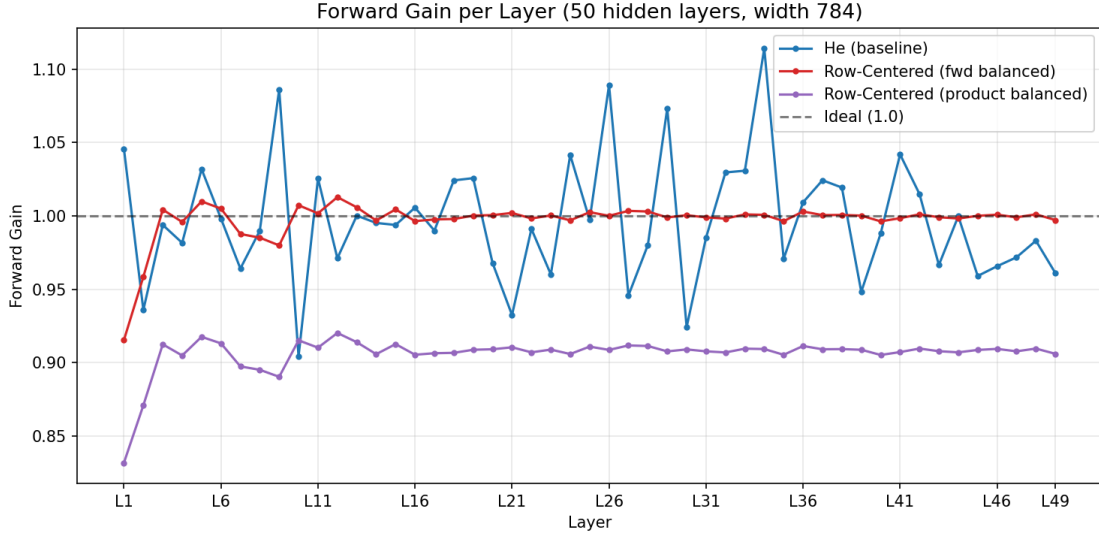


Figure 1: Forward gain per layer. He (blue) fluctuates around 1.0. Fwd-balanced (red) sits precisely at 1.0. Product-balanced (purple) sits at ≈ 0.91 .

4.4 Backward gain per layer

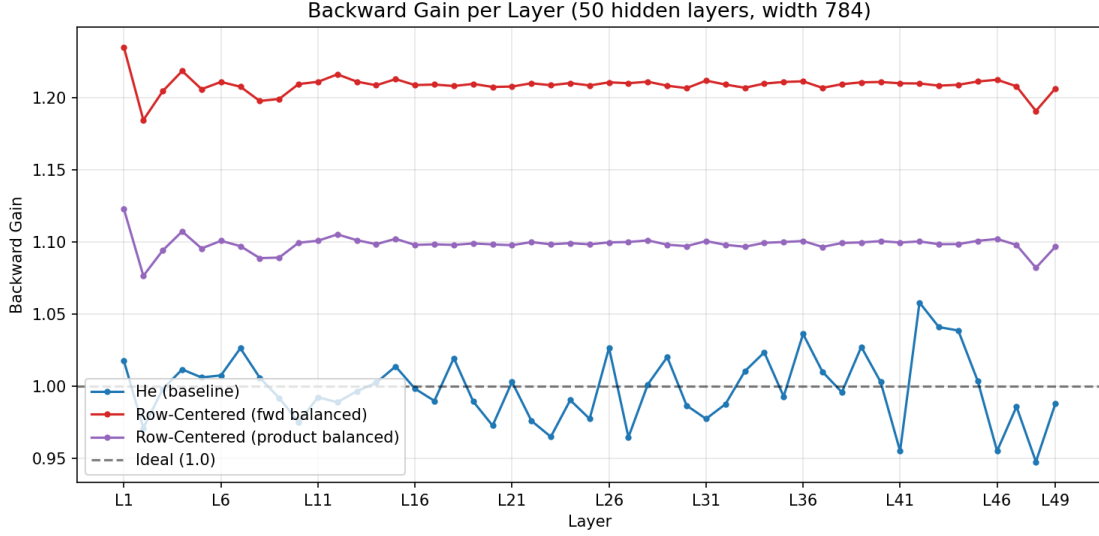


Figure 2: Backward gain per layer. He (blue) fluctuates around 1.0. Product-balanced (purple) sits at ≈ 1.10 . Fwd-balanced (red) is elevated at ≈ 1.21 .

4.5 Gain product per layer

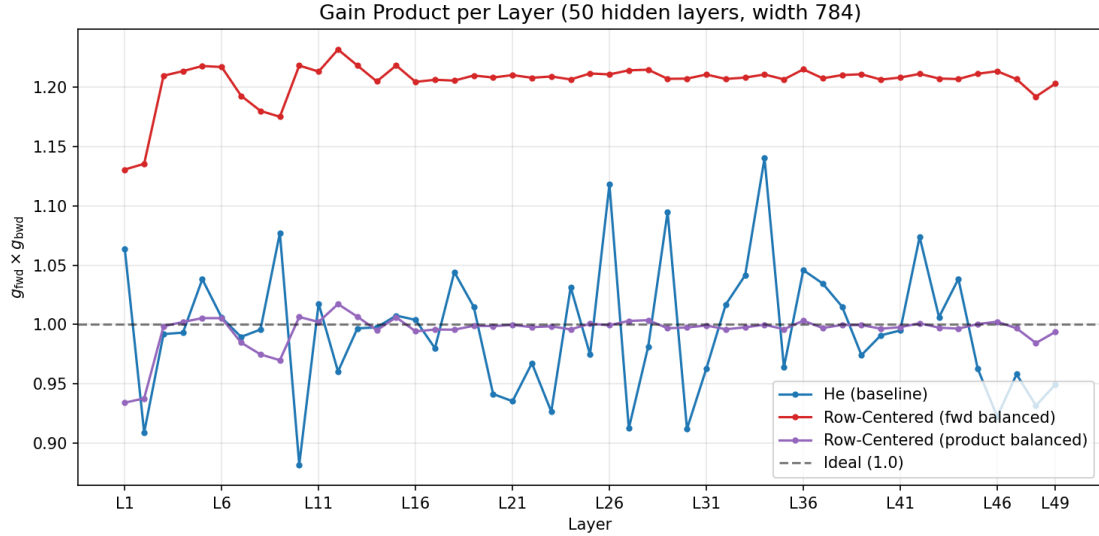


Figure 3: Gain product $g_{\text{fwd}} \times g_{\text{bwd}}$ per layer. **Product-balanced sits exactly at 1.0** (the design target). He sits near 1.0 (both gains individually ≈ 1). Fwd-balanced sits at ≈ 1.21 (product > 1).

5 Geometry Preservation (k-NN Accuracy)

5.1 Setup

Fashion-MNIST, 2000 samples, depths $L \in \{5, 10, 15, 20\}$, square projections (width = 784). The k-NN accuracy ($k = 5$) measures whether class structure survives the multi-layer projection.

5.2 Results

Table 3: k-NN accuracy vs. depth. Chance level = 0.10 (10 classes).

Initializer	5 L	10 L	15 L	20 L
He (baseline)	0.829	0.800	0.784	0.752
RC fwd-balanced	0.810	0.636	0.356	0.270
RC product-balanced	0.810	0.636	0.356	0.269

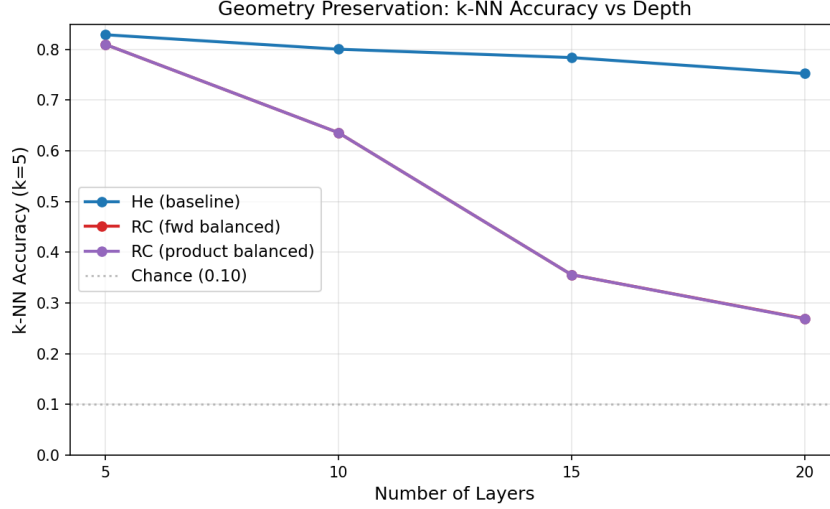


Figure 4: k-NN accuracy vs. depth. He degrades gracefully ($0.83 \rightarrow 0.75$). Both RC variants collapse identically to 0.27 by 20 layers.

Remark 5.1. Product-balanced and fwd-balanced produce identical k-NN curves at every depth. Both are fully row-centered ($\sum_j W_{ij} = 0$), so the DC-blindness — the mechanism that destroys class structure (see main report, Chapter 5) — is identical regardless of variance. Changing s affects gain magnitudes but not the structural constraint.

6 Per-Layer Variance Scaling: The η -Schedule

Since uniform variance cannot produce uniform gradients, we derive a per-layer variance schedule from BP4.

6.1 Derivation from BP4

Step 1: Forward chain. Using (8), the activation RMS at layer $\ell - 1$ is:

$$\text{rms}(A^{(\ell-1)}) = \text{rms}(A^{(0)}) \cdot \prod_{k=1}^{\ell-1} g_{\text{fwd}}^{(k)} = \text{rms}(A^{(0)}) \cdot \prod_{k=1}^{\ell-1} (r \cdot s_k) = \text{rms}(A^{(0)}) \cdot r^{\ell-1} \cdot \prod_{k=1}^{\ell-1} s_k. \quad (16)$$

Step 2: Backward chain. The backward gain at position k is produced by the weight matrix $W^{(k+1)}$, so $g_{\text{bwd}}^{(k)} \approx s_{k+1}$. Telescoping from layer L down to layer ℓ :

$$\text{rms}(\Delta^{(\ell)}) = \text{rms}(\Delta^{(L)}) \cdot \prod_{k=\ell}^{L-1} g_{\text{bwd}}^{(k)} = \text{rms}(\Delta^{(L)}) \cdot \prod_{k=\ell+1}^L s_k. \quad (17)$$

Step 3: Gradient at layer ℓ . Substituting (16) and (17) into (2):

$$\begin{aligned} \left\| \frac{\partial \mathcal{C}}{\partial W^{(\ell)}} \right\| &\propto r^{\ell-1} \cdot \prod_{k=1}^{\ell-1} s_k \cdot \prod_{k=\ell+1}^L s_k \\ &= r^{\ell-1} \cdot \frac{\prod_{k=1}^L s_k}{s_\ell}. \end{aligned} \quad (18)$$

Since $\prod_{k=1}^L s_k$ is a constant (independent of ℓ):

$$\boxed{\left\| \frac{\partial \mathcal{C}}{\partial W^{(\ell)}} \right\| \propto \frac{r^{\ell-1}}{s_\ell}.} \quad (19)$$

Step 4: Condition for uniform gradients.

$$\frac{r^{\ell-1}}{s_\ell} = \text{const for all } \ell \iff s_\ell \propto r^{\ell-1}. \quad (20)$$

6.2 The η -parameterization

Rather than setting $s_\ell = C \cdot r^{\ell-1}$ directly (which would fully correct the gradient at the cost of extreme variance differences between layers), we introduce a *correction strength* parameter $\eta \in [0, 1]$:

$$s_\ell = s_{\text{base}} \cdot r^{\eta(\ell-(L+1)/2)}, \quad (21)$$

where:

- $\ell \in \{1, \dots, L\}$ is the 1-indexed layer number.
- L is the total number of weight layers.
- The exponent is centered at the middle layer $\ell = (L+1)/2$, so the geometric mean of all s_ℓ equals s_{base} .
- $s_{\text{base}} = s^* = (\pi/(\pi-1))^{1/4}$ is the product-balanced factor from Section 3.

The parameter η is **not derived from theory** — it is a tunable knob that controls the trade-off between gradient uniformity and variance uniformity:

- $\eta = 0$: uniform variance ($s_\ell = s^*$ for all ℓ). Gradient ratio $= r^{-(L-1)}$. No correction.
- $\eta = 1$: full correction ($s_\ell \propto r^{\ell-1}$). Gradient ratio $= 1$ (perfect uniformity), but variance ratio $= r^{-(L-1)}$ (extreme).
- $0 < \eta < 1$: partial correction — a compromise.

The per-layer weight standard deviation is then:

$$\sigma_\ell = \sqrt{\frac{2}{d_{\ell-1}}} \cdot s_\ell = \sqrt{\frac{2}{d_{\ell-1}}} \cdot s^* \cdot r^{\eta(\ell-(L+1)/2)}. \quad (22)$$

The initialization procedure is the same as Definition 3.1, but with σ_ℓ varying per layer.

7 The Pareto Frontier

7.1 The fundamental trade-off

Theorem 7.1 (Pareto budget). *For the η -schedule (21), define:*

$$G(\eta) = r^{-(1-\eta)(L-1)} \quad (\text{gradient ratio: max/min gradient}), \quad (23)$$

$$V(\eta) = r^{-\eta(L-1)} \quad (\text{variance ratio: max/min weight std}). \quad (24)$$

Then their product is independent of η :

$$\boxed{G(\eta) \cdot V(\eta) = r^{-(L-1)}.} \quad (25)$$

Proof.

$$\begin{aligned}
G(\eta) \cdot V(\eta) &= r^{-(1-\eta)(L-1)} \cdot r^{-\eta(L-1)} \\
&= r^{-[(1-\eta)+\eta](L-1)} \\
&= r^{-(L-1)}.
\end{aligned}$$

□

This means: reducing the gradient ratio by a factor of k necessarily *increases* the variance ratio by the same factor. The total “budget” $r^{-(L-1)}$ is fixed.

7.2 Numerical consequences

Table 4: Pareto budget at various depths. The budget grows exponentially, making the trade-off increasingly severe.

Depth L	Budget $r^{-(L-1)}$
10	$5.6\times$
20	$38\times$
50	$11,943\times$
100	1.7×10^8

At $L = 20$, the budget is $38\times$ — one can set $\eta = 0.75$ to get gradient ratio $\approx 2.5\times$ and variance ratio $\approx 15\times$, both manageable.

At $L = 50$, the budget is $\sim 12,000\times$. Even the “fairest” split ($\eta = 0.5$, both factors $\approx 109\times$) is severe.

7.3 Depth-adaptive η

Given a target gradient ratio T , one can solve for the required η from (23):

$$\eta^*(L, T) = \max\left(0, 1 - \frac{\ln T}{(L-1) \cdot \ln(1/r)}\right). \quad (26)$$

Table 5: Depth-adaptive η for target gradient ratio $T = 10\times$.

L	η^*	Gradient ratio	Variance ratio
10	0.000	$5.6\times$	$1.0\times$
20	0.368	$10\times$	$3.8\times$
50	0.755	$10\times$	$1,194\times$
100	0.879	$10\times$	$17,276,021\times$

At $L = 50$, achieving gradient ratio $\leq 10\times$ requires variance ratio $\geq 1,194\times$: the weight standard deviation at the first layer would be $\sim 38\times$ larger than at the last layer. While this is numerically representable in float32, the extreme scale difference between layers creates practical training difficulties (e.g., a single learning rate cannot serve all layers well).

7.4 Varying widths

If $d_{\ell-1} \neq d_\ell$, equation (18) acquires an additional factor $\sqrt{d_\ell \cdot d_{\ell-1}}$:

$$\left\| \frac{\partial \mathcal{C}}{\partial W^{(\ell)}} \right\| \propto \sqrt{d_\ell \cdot d_{\ell-1}} \cdot \frac{r^{\ell-1}}{s_\ell}.$$

The $\sqrt{d_\ell \cdot d_{\ell-1}}$ factor is layer-dependent but can be absorbed into the per-layer scaling s_ℓ . The fundamental Pareto structure (Theorem 7.1) is unchanged: the $r^{\ell-1}$ term still requires non-uniform s_ℓ to cancel, and the budget $r^{-(L-1)}$ remains.

8 Conclusion

8.1 Results

1. The product-balanced variance $v^* = 2\sqrt{\pi/(\pi-1)}/d \approx 2.422/d$ is the unique solution to $g_{\text{fwd}} \cdot g_{\text{bwd}} = 1$. Empirically verified to 0.2% accuracy at 50 hidden layers.
2. The Pareto budget theorem: for any row-centered per-layer schedule, $G(\eta) \cdot V(\eta) = r^{-(L-1)}$. This budget grows exponentially with depth and cannot be reduced by any choice of η .
3. Geometry experiments (k-NN accuracy) confirm that *all* row-centered variants — regardless of variance — produce identical geometry degradation. The DC-blindness inherent to row centering is the mechanism, and it is unaffected by s .

8.2 Implications

For deep networks ($L \geq 50$):

- **Gradient stability** within the row-centering framework requires a per-layer variance schedule, but the exponential Pareto budget makes this increasingly impractical with depth. Escaping the Pareto frontier requires abandoning the structural constraint $\sum_j W_{ij} = 0$ — i.e., moving away from full row centering.
- **Geometry preservation** is impossible with row centering at depth. The DC-blindness that defines row centering is the same mechanism that destroys class structure.
- The product-balanced initializer remains useful as an **analytical tool**: it identifies the midpoint of the gain space and provides the natural base for per-layer schemes at moderate depth ($L \leq 20$).